Computing Foundations
6 ECTS
computational_thinking_foundations
computer_operation_and_execution
binary_information_representation
information_representation
computer_software_and_operating_systems
computer_architecture_foundations

Programming Foundations
6 ECTS
imperative_algorithmic_foundations
algorithmic_foundations
general_programming_foundations
programming_language_concepts
procedural_control_and_parameters
recursion

Python Programming
6 ECTS
python_introduction_and_execution
python_core_syntax
python_functions
python_data_structures_and_sequences
python_execution_and_scope
python_object_fundamentals

C Programming
6 ECTS
c_language_core
c_arrays_and_strings
c_functions_and_building
c_input_output_and_files
c_memory_and_pointers
c_data_structures_and_records

C++ Programming
6 ECTS
cpp_programming_fundamentals
cpp_language_basics
cpp_arrays_and_pointers
cpp_memory_and_build_organization
cpp_object_oriented_programming
cpp_object_oriented_and_generic_programming

Data Structures and Algorithms Fundamentals
6 ECTS
abstract_data_types
fundamental_data_structures_and_search
linear_data_structures
dictionary_structures
sorting_data
algorithm_foundations

Advanced Data Structures and Algorithmic Design
3 ECTS
introductory_tree_structures
graph_data_structures
algorithmic_design_techniques

Advanced Algorithms and Complexity
6 ECTS
algorithmic_search_strategies
approximation_and_heuristic_methods
computational_complexity
computational_complexity_2
graph_optimization_algorithms
optimization_search

Logic and Formal Methods
6 ECTS
propositional_logic
first_order_logic
formal_logic_foundations
mathematical_logic_foundations
automated_reasoning
logical_formalisms

Computation and Language Theory
6 ECTS
turing_machines
computability_and_enumerability
computability_limits
regular_languages_and_automata
context_free_languages_and_automata
formal_language_classifications

Programming Languages
6 ECTS
programming_language_description
programming_language_types
programming_language_exceptions
language_implementation_mechanisms
runtime_environments_and_memory
abstract_execution_models

Object-Oriented Software Development
6 ECTS
object_oriented_programming_foundations
object_oriented_programming_foundations_2
object_oriented_software_design
software_design_patterns
software_requirements_analysis
software_lifecycle_activities

Software Engineering
6 ECTS
software_development_methodologies
software_testing_foundations
software_quality_evaluation_and_certification
software_build_and_delivery_automation
software_architecture_design
version_control_fundamentals

Database Systems Fundamentals
5 ECTS
entity_relationship_and_conceptual_design
relational_data_model_and_keys
logical_relational_design
sql_definition_and_querying
formal_relational_query_languages

Database Systems and Internals
4 ECTS
relational_dbms_components
database_architectures
database_transactions
database_storage_structures

Advanced Data Management
6 ECTS
nosql_data_models
nosql_data_models_and_architectures
nosql_database_design_and_architecture
nosql_querying
query_processing_and_optimization
data_platform_architectures

Computer Networks Foundations
4 ECTS
computer_network_fundamentals
networking_foundations
network_architecture_and_layers
link_layer_and_lan_access

Internetworking, Transport, and Network Programming
5 ECTS
internetworking_and_ip
ip_network_operations
tcp_udp_and_congestion
transport_services
network_programming

Distributed Systems Foundations and Programming
4 ECTS
distributed_systems_foundations
distributed_communication
remote_procedure_invocation
message_oriented_middleware

Distributed Algorithms, Consistency, and Coordination
5 ECTS
distributed_algorithms
distributed_consistency_and_replication
distributed_consensus_and_ordering
distributed_coordination_models
distributed_system_scalability

Operating Systems Foundations and Services
4 ECTS
operating_system_foundations
operating_system_filesystems
system_calls
shell_scripting

Operating Systems Processes and Resource Management
5 ECTS
processes_threads
concurrent_synchronization
cpu_scheduling
memory_management
io_management

Computer Architecture Fundamentals
6 ECTS
computer_system_architecture
instruction_set_architecture
processor_organization_and_isa
memory_and_addressing_hardware
processor_io_and_interrupts
architecture_performance_foundations

Advanced Processor Execution and Performance
3 ECTS
computer_performance_measurement
pipelined_processor_design
advanced_instruction_execution

Digital Logic Fundamentals
5 ECTS
electronics_fundamentals_and_instruments
boolean_logic_basics
combinational_logic_design
sequential_logic_basics
synchronous_asynchronous_logic

Embedded Systems and Programming
4 ECTS
embedded_systems_foundations
embedded_bare_metal_programming
arduino_programming
serial_communication_protocols

Artificial Intelligence Foundations: Fundamentals
6 ECTS
ai_introduction
intelligent_systems_and_agents
knowledge_representation
logical_programming
ai_problem_solving
ai_search_strategies

Artificial Intelligence Foundations: Advanced Problem Solving
3 ECTS
constraint_satisfaction
planning_foundations
adversarial_search

Machine Learning Foundations and Core Methods
6 ECTS
machine_learning_foundations
classification_fundamentals
classical_classification_methods
clustering_methods
regression_methods
feature_engineering_and_extraction

Decision Trees, Ensembles, and Model Evaluation
3 ECTS
decision_tree_classification
ensemble_classification
machine_learning_model_evaluation

Deep Learning Fundamentals
4 ECTS
neural_networks_and_deep_learning
neural_network_training_mechanics
deep_learning_training_foundations
deep_learning_training_practices

Deep Learning Architectures and Models
5 ECTS
cnn_components_architectures
recurrent_neural_networks
transformer_architectures_and_attention
attention_and_vision_transformers
generative_model_foundations

Data Science Foundations and Analysis
6 ECTS
data_science_foundations_and_workflow
data_analytics_foundations
data_integrity_and_analysis_validity
descriptive_and_inferential_statistics
statistical_inference_and_testing
data_visualization

Data Mining Processes and Validation
3 ECTS
data_mining_process
data_mining_workflows_and_preparation
data_mining_validation

Computer Vision Foundations
6 ECTS
computer_vision_foundations
image_acquisition_sampling
digital_image_representation
image_convolution_filtering_edges
binary_image_analysis
image_classification_representations

Computer Vision Scene Analysis
3 ECTS
image_segmentation_basics
cnn_object_detection
visual_object_analysis

Natural Language Processing Foundations and Text Analysis
5 ECTS
natural_language_processing_foundations
text_tokenization
text_mining_foundations_and_tools
corpus_statistics_and_association
text_retrieval_and_semantic_analysis

Natural Language Processing Models and Applications
4 ECTS
text_classification_models
word_embeddings
opinion_mining
language_model_generation_and_prompting

Cybersecurity Fundamentals and Protection
6 ECTS
cybersecurity_fundamentals
it_security_fundamentals
authentication_and_authorization
cryptographic_security_goals
network_and_communications_protection
system_and_data_protection

Cybersecurity Application, Detection, and Governance
3 ECTS
application_security
cybersecurity_detection_and_response
cybersecurity_governance_and_risk

Cryptographic Foundations and Algorithms
4 ECTS
cryptography_foundations
symmetric_cryptography
public_key_cryptography
cryptographic_hashes

Cryptographic Protocols, Trust, and Security
5 ECTS
cryptographic_protocols
cryptographic_key_management
cryptographic_trust_infrastructure
digital_signatures
formal_protocol_security

Web Programming Foundations
3 ECTS
internet_and_web_foundations
web_protocol_and_uri_foundations
web_markup_and_content_formats

Web Application Development
6 ECTS
client_side_web_programming
client_side_web_frameworks
event_driven_server_programming
server_side_web_programming
rest_web_apis
web_application_architecture_and_state

Cloud Computing Platforms
4 ECTS
cloud_computing_foundations
cloud_infrastructure_and_service_models
cloud_service_orchestration
container_virtualization

Big Data Processing
5 ECTS
big_data_context_and_management
big_data_processing_architectures
mapreduce_programming
spark_big_data_processing
online_stream_processing

Mobile Application Development Foundations and Android
6 ECTS
mobile_application_types
mobile_application_design
mobile_architecture_patterns
android_application_foundations
android_classic_user_interfaces
android_data_management

iOS Application Development
3 ECTS
ios_application_foundations
ios_uikit_interaction
ios_swiftui

Human-Computer Interaction
6 ECTS
interaction_and_interface_foundations
user_experience_foundations
interaction_ideation_and_prototyping
user_research_methods
interactive_system_evaluation
usability_design_and_evaluation

Real-Time Systems Fundamentals
2 ECTS
real_time_foundations
real_time_scheduling

Advanced Real-Time Systems
3 ECTS
aperiodic_scheduling
priority_resource_access
real_time_operating_system_concepts

Concurrency Models and Formalisms
2 ECTS
concurrent_message_passing
concurrent_systems_formalisms

Concurrent Programming and Verification
2 ECTS
shared_memory_concurrency_programming
concurrent_program_model_checking

Blockchain and Cryptocurrencies
6 ECTS
blockchain_and_cryptocurrency_foundations
blockchain_principles
blockchain_network_organization
blockchain_platforms
blockchain_operations_and_contracts
blockchain_security_and_anonymity

Constraint Programming and Scheduling
3 ECTS
constraint_programming_modeling
constraint_search_and_optimization
constraint_based_scheduling

SAT and Satisfiability Modulo Theories
3 ECTS
sat_foundations_and_encodings
advanced_sat_solving
satisfiability_modulo_theories

Linear, Integer, and Combinatorial Optimization
3 ECTS
linear_optimization
integer_linear_programming
combinatorial_optimization_foundations

Reliable Service Engineering Fundamentals
6 ECTS
scalable_reliable_service_architecture
service_performance_capacity_testing
service_observation_and_outage_analysis
service_operational_response
service_organization_and_complexity
cloud_service_deployment_and_migration

Reliable Service Governance and Maturity
3 ECTS
service_maturity_assessment
service_cost_benefit_analysis
data_strategy_and_governance

Virtualized Systems Administration Foundations
5 ECTS
user_level_operating_systems
linux_network_configuration
linux_process_and_resource_monitoring
unix_accounts_and_filesystem_authorization
system_logging_and_scheduled_automation

Virtualized Systems Administration Automation and Monitoring
4 ECTS
docker_resource_management_api
configuration_as_code
centralized_configuration_management
centralized_monitoring

Modern Software Architecture Foundations
6 ECTS
uml_modeling_foundations_and_use_cases
domain_modeling
domain_driven_design
software_architecture_styles
hexagonal_architecture
software_frameworks

Modern Software Architecture Service Systems
3 ECTS
microservice_architectures
service_oriented_and_web_services
web_api_documentation

Advanced Computer Vision: Geometry and Local Methods
3 ECTS
camera_geometry_calibration
local_visual_features
shape_matching_detection

Advanced Computer Vision: Visual Understanding
3 ECTS
advanced_image_segmentation
depth_estimation
visual_metric_learning

Advanced Computer Vision: Generation and Robustness
3 ECTS
diffusion_generation_adaptation
generative_model_evaluation
adversarial_neural_inputs

Scientific Python and Data Foundations
3 ECTS
python_programming_basics
python_file_handling
python_text_processing

Scientific Python and Data Computing
6 ECTS
python_modules_dependencies_and_packaging
python_tabular_and_spreadsheet_data
python_scientific_and_numerical_computing
python_numerical_computing
python_data_analysis_libraries
pandas_tabular_data

Mobile Systems and Middleware Fundamentals
6 ECTS
mobile_middleware
mobile_execution_fundamentals
mobile_networking_services
mobile_data_synchronization
mobile_messaging
mobile_multimedia

Mobile Systems and Middleware Application Capabilities
3 ECTS
mobile_debugging
mobile_pervasive_computing
mobile_positioning

Digital Health Systems: Foundations and Applications
6 ECTS
digital_health_foundations
healthcare_delivery_and_payer_models
digital_health_service_improvement
digital_health_platforms_and_connected_systems
personal_health_informatics_and_systems
mobile_health_applications

Digital Health Systems: Interventions, Evaluation, and Governance
3 ECTS
digital_therapeutics_and_behavior_change
digital_health_outcomes_evaluation
digital_health_clinical_research_governance

Digital Resources and Digitization
6 ECTS
digital_humanities_data_management
digital_resource_design
digital_textualization_and_analysis
digital_image_quality_and_restoration
digital_image_management_and_preservation
document_digitization_acquisition

Open Science Practices
3 ECTS
open_science_foundations_and_practices
open_data_software_and_reproducibility
semantic_publishing

Vulnerability Assessment and Offensive Security
3 ECTS
vulnerability_identification
vulnerability_assessment_and_exploitation
offensive_security_methods

Advanced Cybersecurity Approaches and Cryptography
3 ECTS
advanced_cybersecurity_approaches
cryptographic_randomness
post_quantum_cryptography

Federated Authentication and Cryptographic Communication
3 ECTS
federated_authentication
federated_authentication_protocols
pgp_cryptographic_communication

Large Language Models and Applications
6 ECTS
large_language_model_fundamentals
large_language_model_training_and_adaptation
large_language_model_scaling
efficient_and_compressed_neural_architectures
agents_actions_and_function_calls
retrieval_augmented_generation_and_vector_databases

Graph and Neurosymbolic Knowledge Systems
3 ECTS
graph_machine_learning
knowledge_graph_embeddings_and_language_models
neurosymbolic_methods

Digital Forensics
6 ECTS
digital_forensics_foundations_and_practice
forensic_data_acquisition_and_evidence_handling
forensic_analysis_reporting_and_testimony
specialized_digital_forensics_investigations
forensic_systems_and_file_recovery
anti_forensics

Parallel Computer Hardware and Architectures
6 ECTS
memory_hierarchy_and_caches
intel_memory_management_and_protection
shared_memory_multiprocessor_architectures
computer_interconnects_and_dma_io
processor_hardware_virtualization
parallel_architectures

Parallel Computation Foundations and Programming
3 ECTS
parallel_computation_foundations
parallel_performance_and_scalability
gpu_and_vector_parallel_programming

Network Communication Foundations
3 ECTS
physical_layer_transmission
network_switching
channel_reliability

Internet Services and Protocol Analysis
3 ECTS
application_layer_protocols
dns_and_name_systems
network_protocol_analysis

Wireless Networking and Mobility
3 ECTS
wireless_radio_systems
wireless_networking_technologies
transport_and_network_service_quality

Formal Language and Language Analysis Tools
6 ECTS
formal_syntax_notation
formal_parsing_methods
lexical_analysis
lex_tooling
parser_generation_tooling
antlr_language_tools

Compiler Construction and Static Analysis
3 ECTS
program_syntax_analysis
compiler_intermediate_and_target_code
compiler_semantic_and_static_analysis

Business Intelligence and Data Warehousing
6 ECTS
business_intelligence_foundations
data_warehouse_design
data_warehouse_etl
bi_reporting_and_sharing
bi_visualization_tools
r_for_business_analytics

IoT Time-Series and Sensor Forecasting
3 ECTS
statistical_time_series_forecasting
iot_time_series_data
ai_sensor_forecasting

Agricultural IoT Foundations, Networks, and Platforms
6 ECTS
iot_foundations
iot_platform_environments
embedded_and_wireless_networks
iot_wireless_technologies
iot_messaging_protocols
iot_network_routing

Agricultural Embedded IoT Programming and Edge AI
3 ECTS
embedded_iot_programming
ai_ml_for_big_data
edge_ai_and_tinyml

Simulation Systems
5 ECTS
simulation_concepts
simulation_modeling_methods
simulation_implementation
simulation_study_evaluation
simulation_and_machine_learning

Immersive Reality Systems
4 ECTS
immersive_reality_introduction
mixed_reality_foundations
immersive_software_platforms
immersive_hardware_and_human_factors

Software Quality Foundations
6 ECTS
command_line_interaction
distributed_version_control
code_quality_and_refactoring
software_test_practices
software_testing_specification
software_performance_engineering

Software Release Management
3 ECTS
software_release_processes
software_versioning_and_commit_conventions
software_licensing

Agentic Systems and Reinforcement Learning
5 ECTS
agent_concepts
agentic_ai_systems
reinforcement_learning_foundations
reinforcement_learning_methods
multi_agent_reinforcement_learning

AI Alignment, Safety, and Computational Creativity
4 ECTS
ai_epistemology_and_ethics
ai_safety_and_alignment
generative_ai_and_creativity
preference_based_ai_alignment

Quantum Computing
6 ECTS
quantum_computing_foundations
quantum_circuit_models
quantum_programming
quantum_algorithms
quantum_complexity
quantum_security_and_error_correction

Service Delivery
2 ECTS
service_lifecycle
secure_service_delivery

Virtual Systems
5 ECTS
virtuality_fundamentals
virtual_machine_implementation
virtual_network_implementation
virtual_networking_stacks
virtual_devices_and_file_systems

Distributed Storage
2 ECTS
distributed_file_systems
distributed_block_storage

Identity Management and Systems Integration
4 ECTS
identity_management
ldap_authentication
kerberos_single_sign_on
windows_linux_integration

Network Communication and Security
5 ECTS
network_connection_monitoring
secure_remote_administration
datagram_transport_security
nat_traversal
proxy_protocols

UML Modeling and Diagram Generation
5 ECTS
uml_structural_modeling
uml_behavior_modeling
uml_interaction_modeling
uml_architecture_modeling
uml_diagram_generation

Industrial Software Communication
2 ECTS
dotnet_and_csharp
opc_industrial_communication

Distributed Systems and Multi-Agent Coordination
2 ECTS
distributed_operating_systems
multi_agent_systems_and_coordination

Visual Geometry and Feature Foundations
3 ECTS
camera_geometry_calibration
local_visual_features
shape_matching_detection

Neural Visual Learning Methods
6 ECTS
neural_network_visualization
neural_image_transformation
visual_sequence_captioning
vision_language_models
advanced_image_segmentation
diffusion_generation_adaptation

Concurrent Formal Models Fundamentals
3 ECTS
concurrent_system_models
concurrent_process_algebra
ccs_and_structural_semantics

Concurrent Formal Model Equivalences and Properties
6 ECTS
behavioral_bisimulation
petri_net_equivalences
concurrency_modal_logics
distributed_computability
temporal_concurrency_properties
temporal_system_specification

Python Programming Foundations
3 ECTS
scripting_and_generic_text_processing
python_turtle_graphics
programming_classes

Data Management and Pipeline Design
3 ECTS
json_data_representation
database_management_systems
data_pipeline_design

Python Data Analytics
3 ECTS
descriptive_data_analytics
probabilistic_and_stochastic_computation
predictive_modeling

Biological Data and Sequence Algorithms
5 ECTS
biological_data_representation
scientific_data_formats
sequence_alignment_heuristics
sequence_alignment_and_distance
biological_sequence_algorithms

Numerical Computation and Pattern Recognition
4 ECTS
numerical_computation_algorithms
numerical_approximation_methods
numerical_error_analysis
computational_pattern_recognition

Enterprise Information Systems and Digital Transformation
6 ECTS
digital_transformation_concepts
enterprise_information_system_categories
information_system_portfolio_and_planning
information_system_data_integration
business_process_reengineering
initiative_feasibility_and_business_planning

Model-Driven and Multiplatform Application Development
3 ECTS
model_driven_software_development
kotlin_programming_and_multiplatform
spring_boot_application_development

Concurrency Programming Fundamentals
3 ECTS
java_concurrent_programming
asynchronous_programming
channel_message_passing

Advanced Concurrent Programming
2 ECTS
actor_concurrency
temporal_concurrency_properties

Mobile Applications and Data Access
4 ECTS
financial_data_access
hybrid_mobile_applications
cross_platform_mobile_development
mobile_execution_fundamentals

Android Development
7 ECTS
android_development_setup
kotlin_android_programming
android_compose_ui
android_navigation
android_runtime_execution
android_camera
mobility_handoff_management

iOS Development
2 ECTS
ios_development_setup
swift_ios_programming

Data Management and Service Integration
4 ECTS
ios_core_data
enterprise_messaging
event_publish_subscribe
service_discovery

Wireless Networks and Access Systems
5 ECTS
cellular_networks_handoff
manet_routing
open_ran
wireless_access_standards
zigbee_networks

Mobile and Wireless Network Foundations
3 ECTS
wireless_networking_technologies
mobile_ip_architectures
virtual_channels_and_networks

Network Communication Services
4 ECTS
sip_ims
voip_telephone_systems
network_service_architectures_and_applications
group_communication

Network Management and Troubleshooting
2 ECTS
network_management_and_monitoring
network_troubleshooting

Python Development and Data Communication
3 ECTS
python_development_environments
python_data_visualization
data_storytelling

Machine Learning and Applied Analytics Workflows
3 ECTS
machine_learning_workflows
time_series_analysis
data_science_fashion_applications

Intelligent Systems Methods and Applications
3 ECTS
systematic_literature_review
reinforcement_learning
human_activity_recognition

Digital Culture and Intelligence Foundations
3 ECTS
digital_and_analog_representations
digitalization_impacts
digital_data_and_intelligence

Digital Decision Systems and Applications
6 ECTS
digital_memory_architectures
automatic_decision_systems
digital_decision_making_and_forecasting
digital_text_analysis
generative_ai_platforms
business_process_agents

Digital Health Foundations and Applications
4 ECTS
iot_sensing_and_smart_sensors
medical_devices_and_regulation
health_informatics_introduction
digital_health_case_studies

Clinical Information and Neurophysiology Systems
5 ECTS
medical_document_lifecycle
clinical_data_archiving
neurophysiology_information_systems
eeg_acquisition_and_storage
digital_signal_processing

Making and Digital Fabrication
2 ECTS
making_problem_solving
digital_fabrication_and_3d_printing

Digital Electronics and Motor Control
3 ECTS
visual_block_programming
verilog_digital_design
motor_control_circuits

Internet of Things Systems and Smart Spaces
4 ECTS
iot_architectures
iot_application_models_and_mechanisms
iot_entity_identification
smart_spaces

Computational Linguistics Foundations
2 ECTS
computational_linguistics_foundations
natural_language_grammar_and_parsing

Computer-Assisted Translation Foundations and Workflows
2 ECTS
computer_assisted_translation_foundations
cat_workflow_management

Machine Translation and Translation AI Services
5 ECTS
cat_machine_translation_integration
machine_translation_foundations
machine_translation_operations
translation_revision_and_evaluation
translation_ai_ethics

Cybersecurity Contexts and Personal Security
3 ECTS
cybersecurity_contexts
personal_and_internet_security
digital_rights_and_law

Malware, Network Attacks, and Attack Analysis
6 ECTS
malicious_software
reverse_engineering
denial_of_service_attacks
wifi_network_security
software_defined_radio_attacks
tor_and_dark_web

Digital Graphics and Media Foundations
6 ECTS
data_compression
digital_graphics_foundations
vector_graphics
digital_content_production
web_resource_distribution_frameworks
digital_photography

Computer-Aided Landscape Representation
3 ECTS
computer_aided_design
autocad_2d_drawing
landscape_project_representation

Digital Acquisition and Point-Cloud Processing
3 ECTS
archaeological_digital_capture
heritage_3d_acquisition
point_cloud_processing

Three-Dimensional Modeling and Virtual Reconstruction
3 ECTS
three_d_modeling
virtual_restoration
archaeological_virtual_reconstruction

Cultural Heritage Communication and Digital Presentation
3 ECTS
cultural_heritage_communication
heritage_web_information_systems
heritage_three_d_printing

Data Mining Platforms
3 ECTS
data_mining_infrastructure
data_warehouse_design_and_olap
data_science_development_tools

Data Mining Methods and Applications
6 ECTS
outlier_detection
bayesian_network_classification
weka_data_mining_practice
spark_data_processing_and_machine_learning
text_mining_applications
association_rule_mining

Large Language Model Foundations and Applications
6 ECTS
large_language_model_fundamentals
large_language_model_training_and_adaptation
large_language_model_scaling
efficient_and_compressed_neural_architectures
agents_actions_and_function_calls
retrieval_augmented_generation_and_vector_databases

Structured Knowledge for Language Models
3 ECTS
graph_machine_learning
knowledge_graph_embeddings_and_language_models
text_knowledge_extraction

Digital Forensics and Investigation Foundations
3 ECTS
digital_forensics_foundations_and_practice
forensic_data_acquisition_and_evidence_handling
forensic_analysis_reporting_and_testimony

Advanced Digital Forensics Investigations
3 ECTS
specialized_digital_forensics_investigations
forensic_systems_and_file_recovery
anti_forensics

Digital Investigation Security Techniques
3 ECTS
vulnerability_assessment_and_exploitation
reverse_engineering
digital_signatures

Computer Architecture Extensions Fundamentals
4 ECTS
numeric_representation
machine_languages
array_algorithms
high_level_assembly_mapping

Advanced Computer Architecture Extensions
5 ECTS
memory_hierarchy_and_caches
intel_memory_management_and_protection
shared_memory_multiprocessor_architectures
computer_interconnects_and_dma_io
processor_hardware_virtualization

Project Management
6 ECTS
information_system_and_project_risk_analysis
project_management_foundations
project_scope_definition
project_planning_and_control
project_stakeholder_communications
project_procurement_management

Digital Innovation
3 ECTS
digital_business_ideation_and_diagnosis
digital_innovation_collaboration
digital_innovation_deployment_and_followup

Data Structures
6 ECTS
binary_tree_structures
queue_programming
set_structures
searching_data
heap_priority_structures
balanced_search_trees

Algorithm Design and String Processing
3 ECTS
recursive_algorithm_design
sorting_algorithms_and_bounds
string_matching_and_indexing

C Data Structures and Trees
3 ECTS
c_linear_data_structures
c_trees
cpp_tree_structures

Graph and Sequence Algorithms
6 ECTS
graph_search
sequence_alignment_and_distance
biological_sequence_algorithms
algorithmic_search_strategies
graph_optimization_algorithms
array_algorithms

Formal Logic and Computational Foundations
5 ECTS
modal_and_epistemic_logic
axiomatic_set_theory
logical_paradoxes
structural_reasoning_and_recursion
computational_complexity_classes

AI Reasoning and Planning
4 ECTS
planning_methods
probabilistic_reasoning
rule_inference
prolog_programming

Systems and Network Services Foundations
3 ECTS
interprocess_communication
physical_layer_transmission
transport_and_network_service_quality

Advanced Network Services and Communication
6 ECTS
network_security_protocols
network_service_architectures_and_applications
voip_telephone_systems
group_communication
mobile_ip_architectures
network_troubleshooting

Computer Architecture and Processor Systems
5 ECTS
computer_architecture_history
combinational_logic_circuits
sequential_logic_circuits
microprocessor_system_design
computer_storage

Operating-System Implementation and Real-Time Programming
4 ECTS
secondary_storage
os_security
linux_kernel
vxworks_realtime_programming

Web Application Architecture Fundamentals
3 ECTS
web_application_fundamentals
web_architectures_and_stacks
web_protocol_evolution

Advanced Web Application Architecture
6 ECTS
single_page_front_end_applications
model_based_web_application_design
web_persistence_and_database_access
enterprise_containers_and_components
enterprise_runtime_services
function_as_a_service

Information Representation and Document Foundations
3 ECTS
information_representation_basics
document_file_formats
character_encodings

Digital Document and XML Processing
2 ECTS
digital_document_management
xml_processing_and_transformation

Information Retrieval
2 ECTS
information_retrieval_foundations
web_information_retrieval

Visual Programming and Digital Humanities Projects
2 ECTS
visual_programming
digital_humanities_projects

Computation Foundations
3 ECTS
programming_language_history
computational_iteration_and_recursion
lambda_calculus

Program Formalization
2 ECTS
program_syntax_analysis
structured_operational_semantics

Concurrent and Advanced Systems
4 ECTS
oracle_computation
distributed_computability
concurrent_system_models
temporal_system_specification

Linguistic and Statistical Foundations of Natural Language Processing
6 ECTS
computational_morphology
lexical_semantics_and_word_senses
logical_sentence_semantics
ngram_language_models
named_entity_and_part_of_speech_annotation
text_similarity_and_dissimilarity

Contextual and Neural Methods in Natural Language Processing
3 ECTS
contextual_word_embeddings
text_convolutional_networks
language_model_prompting

Speech and Text Analytics
3 ECTS
speech_phonetics_and_processing
neural_speech_recognition
topic_modeling

Knowledge Graphs and Semantic Web
6 ECTS
knowledge_graph_foundations
ontology_representation_and_engineering
semantic_web_query_and_search
semantic_web_logical_inference
knowledge_graph_integration_and_completion
knowledge_graph_programming

Knowledge Graph Analysis and Visualization
1 ECTS
knowledge_graph_analysis_and_visualization

Web Development Foundations
4 ECTS
javascript_programming
javascript_prototype_objects
web_language_supersets
web_build_tools

Server-Side Web Development
4 ECTS
nodejs_web_development
python_server_side_web_development
asynchronous_web_workflows
web_application_architectural_patterns

Excel Data Preparation and VBA Automation
3 ECTS
excel_data_preparation
vba_macro_automation
ai_assisted_vba

R Programming Foundations
2 ECTS
r_programming_foundations
r_data_structures

R Data Tools
3 ECTS
r_file_input_output
r_random_values
r_graphics_and_packages

Classification Data Analysis
1 ECTS
classification_data_analysis

Database Foundations and Application Access
6 ECTS
data_flow_modelling
database_fundamentals_and_operations
database_management_systems
microsoft_access_databases
sql_access_control
database_application_access

Advanced Database Operations and Queries
3 ECTS
physical_database_storage
relational_database_operations
specialized_multidimensional_queries

Terminology Foundations and Specialised Corpora
2 ECTS
terminology_theory
specialized_corpus_construction

Terminology Extraction and Database Management
2 ECTS
terminology_extraction_and_corpus_analysis
terminology_database_management

Interpreter Documentation and Speech Technology
2 ECTS
interpreter_documentation
speech_recognition_and_live_subtitling

Professional Communication and Research Tools
3 ECTS
professional_email
bibliography_management
search_engines

Statistical Data Foundations
4 ECTS
descriptive_statistics
study_designs
excel_statistical_data_processing
human_data_quality_and_interpretation

Predictive and Decision Analysis
5 ECTS
deductive_inference
predictive_modeling
probabilistic_and_stochastic_computation
statistical_time_series_forecasting
algorithmic_decision_oversight

Multimedia Systems
5 ECTS
multimedia_information_representation
multimedia_database_management
multimedia_retrieval
multimedia_exploration_and_visualization
multimedia_semantics_and_recommendation

Social and Web Information
4 ECTS
documentary_information_services
social_networks
semantic_web_and_web_of_data
search_engines

Digital Administration and Public Services
5 ECTS
information_systems_basics
electronic_documents_and_trust_services
digital_transformation_legislation
italian_government_portals
cloud_and_web_collaboration

Healthcare Information Systems
4 ECTS
italian_health_system
healthcare_information_system_operations
healthcare_data_and_records
healthcare_activity_classification

Distributed Systems and Cyber-Physical Simulation
3 ECTS
distributed_dependability_and_qos
distributed_data_and_code_technologies
distributed_cyber_physical_simulation

Concurrent and Formal System Modelling
2 ECTS
concurrent_process_algebra
formal_system_modelling

Model Checking
2 ECTS
model_checking

AI-Assisted Software Development and Testing
2 ECTS
ai_assisted_software_development
software_testing_specification

Functional Effects and Monads
2 ECTS
functional_effects_and_monads

Adaptive Systems and Multi-Agent Foundations
6 ECTS
markov_process_models
self_organizing_adaptive_systems
information_theory_fundamentals
multi_agent_systems_and_coordination
agent_concepts
systematic_literature_review

Reinforcement Learning Systems
3 ECTS
reinforcement_learning
reinforcement_learning_applications
multi_agent_reinforcement_learning

Sensor Measurement and Analysis
4 ECTS
sensor_measurement_and_modeling
sensor_acquisition_and_signal_quality
sensor_data_analysis_and_reduction
sensor_feedback_and_tradeoffs

Embedded Networks and Interfaces
5 ECTS
embedded_sensor_networks
embedded_and_wireless_networks
iot_sensing_and_smart_sensors
serial_communication_protocols
embedded_device_drivers

Software Architecture, Design, and Governance
6 ECTS
software_architecture_foundations
software_architecture_patterns_and_evolution
cognitive_architectures
software_documentation
software_frameworks
software_licensing

Software Development and Integration Practice
3 ECTS
scrum

prolog_java_integration
software_debugging_and_assertions

Web Application Foundations and Technology
6 ECTS
web_browser_server_foundations
javascript_across_web_applications
web_application_technology_stacks
web_build_tools
web_language_supersets
web_application_architectural_patterns

Web Human-Computer Interaction and User Experience
3 ECTS
web_human_computer_interaction
web_usability_and_ux
user_research_and_design_framing

Data Models and Information Retrieval
4 ECTS
data_model_fundamentals
sql_xml_xquery
information_retrieval_foundations
pagerank_web_retrieval

Network Analysis
5 ECTS
network_analysis_foundations
network_data_and_research_methods
network_level_metrics
network_node_metrics
network_visualization

Service Process Modeling and Choreographies
2 ECTS
business_process_modeling
service_choreographies

Service Communication and Distributed Operating Systems
2 ECTS
service_communication_technologies
distributed_operating_systems

Foundations of AI Software Development
2 ECTS
ai_software_development_tools
software_development_cognitive_models

AI Systems, Data Analytics, and Production Models
3 ECTS
software_engineering_for_ai_systems
data_analytics_platforms
ai_blockchain_crypto_production_models

Embedded Real-Time Programming
6 ECTS
general_operating_system_programming
embedded_build_and_porting
real_time_operating_system_implementations
freertos_and_esp32
embedded_device_drivers
event_driven_architectures

Cyber-Physical Systems Integration and Security
3 ECTS
cyber_physical_system_middleware
cyber_physical_system_security
embedded_mobile_integration

Vehicular, Connected, and Digital-Twin Foundations
6 ECTS
digital_twin_simulation
autonomous_vehicle_systems
vehicular_application_scenarios
connected_vehicular_systems
vehicular_communication_conditions
industrial_iot_applications

Autonomous Vehicular Perception, Control, and Simulation
3 ECTS
autonomous_vehicle_perception
autonomous_vehicle_decision_control
carla_vehicular_simulation

Business Analytics and Decision Support
5 ECTS
business_decision_foundations
business_modeling_and_exploration
business_decision_support
business_machine_learning_tools
operational_performance_monitoring

Data Governance and Sustainable Business Technologies
4 ECTS
data_strategy_and_governance
esg_data_and_monitoring
e_governance_business_tools
sustainable_information_technologies

Fashion E-Commerce, Web, and Mobile Foundations
5 ECTS
ecommerce_platform_foundations
fashion_ecommerce_analysis
mobile_platforms_and_technology_evolution
web_mobile_architectures
web_mobile_development_frameworks_and_tools

Immersive Fashion Commerce Applications
4 ECTS
immersive_fashion_ecommerce
immersive_interaction_design
immersive_content_creation
immersive_environments_and_applications

Computer Science Education Foundations and Methods
6 ECTS
computer_science_concepts_and_context
computer_science_learning_theories
computer_science_curricula
computer_science_teaching_methods
making_and_unplugged_cs_education
making_problem_solving

Programming and Software in Computer Science Education
3 ECTS
programming_education
programming_learning_activities
educational_software_licensing

Functional Programming Foundations and Scala
2 ECTS
functional_programming_foundations
functional_programming_scala

Object-Oriented Models, Relationships, and Reusable Behavior
3 ECTS
object_oriented_programming_models
object_oriented_relationships
language_type_classes_and_traits

Advanced Types, Effects, Certification, and Concurrency
4 ECTS
advanced_type_systems
functional_effect_composition
certified_programming
software_transactional_memory

Quantum Computation
6 ECTS
quantum_computing_foundations
quantum_circuit_models
quantum_programming
quantum_algorithms
quantum_complexity
quantum_security_and_error_correction

Computational Learning and Complexity
3 ECTS
computational_learning_theory
polynomial_time_computability
oracle_computation

Robotics Foundations and Autonomous Robot Behavior
6 ECTS
robotics_foundations
autonomous_mobile_robotics
robot_navigation_and_planning
robot_behavior_architectures
robot_learning_and_adaptation
robot_control_evaluation

Advanced Robot Design and Swarm Intelligence
3 ECTS
advanced_robot_design_and_control
swarm_robotics
swarm_optimization

AI Foundations of Cognition and Representation
3 ECTS
ai_representation_and_problem_solving
ai_cognition_and_interaction
embodied_and_social_cognition

Meaning and Multimodal Interpretation in AI
3 ECTS
semiotic_and_conceptual_meaning
multimodal_semantic_interpretation
semantic_universals

Computational Methods and Explainability in AI
3 ECTS
neural_networks
bayesian_machine_learning
explainable_artificial_intelligence

Programming and Array Foundations
2 ECTS
introductory_programming_methods
numpy_array_objects

Logic and Rule Programming
4 ECTS
logic_inference_techniques
prolog_programming
rule_inference
inductive_logic_programming

Constraint Programming and Probabilistic Reasoning
3 ECTS
constraint_logic_programming
minizinc_constraint_modeling
probabilistic_reasoning

Scientific Programming and Data Representation
6 ECTS
matlab_programming
python_numerical_computing
vectors_and_tensors
scientific_data_formats
array_algorithms
numeric_representation

Numerical Computation Methods and Error Analysis
3 ECTS
numerical_computation_algorithms
numerical_approximation_methods
numerical_error_analysis

Localization and Conversational Systems
6 ECTS
localization_domains_and_text
localization_production_and_quality
conversational_informatics
language_model_prompting
speech_phonetics_and_processing
neural_speech_recognition

Programmable and Virtualized Networking
5 ECTS
network_switching
programmable_networking
software_defined_and_virtualized_networking
virtual_networking_stacks
virtual_network_implementation

Wireless and Mobile Networks
4 ECTS
wireless_access_standards
zigbee_networks
open_ran
manet_routing

Digital Media Economics and Industries
3 ECTS
digital_media_economics
digital_media_industry_transformation
music_and_video_industry_value_chains

Industrial Digital Transformation and Applications
6 ECTS
industry_transformation_context
industrial_monitoring_and_diagnostics
remaining_useful_life_and_maintenance
optimization_and_machine_learning_integration
probabilistic_and_fair_machine_learning_applications
industrial_decision_applications

User Experience Design Foundations
4 ECTS
mobile_usability_and_ux
web_usability_and_ux
user_models_and_experience_mapping
service_concept_design

Advanced Interaction and Cultural Heritage Experiences
5 ECTS
ux_for_complex_systems
interaction_paradigms
interactive_storytelling
cultural_heritage_interaction_design
virtual_heritage_tours

Visual Media Fundamentals for Cultural Heritage
6 ECTS
digital_graphics_foundations
digital_photography
digital_images_for_cultural_heritage
classic_image_processing_and_representation
image_based_modeling
three_d_modeling

Interactive Cultural Heritage Media Applications
3 ECTS
cultural_heritage_interaction_design
virtual_heritage_tours
heritage_web_information_systems

Geospatial Foundations and Analysis
4 ECTS
gis_fundamentals
cartographic_representation
geospatial_data_management
geospatial_analysis

Context-Aware Systems and Applications
5 ECTS
context_aware_systems
location_aware_services
mobile_edge_service_management
supervised_activity_recognition
differential_drive_robot_applications

Digital Productivity and Research Tools Fundamentals
5 ECTS
informatics_foundations
linux_operating_systems
internet_applications
windows_file_management
office_productivity_basics

Digital Productivity and Research Workflows
4 ECTS
spreadsheet_data_processing_and_reporting
spreadsheet_data_exchange
research_data_presentations
latex_document_authoring

Java and Modular Programming
9 ECTS
modular_programming
java_programming_foundations
java_data_structures
java_language_features
java_input_output
java_graphical_and_event_programming
java_functional_programming
latex_collaborative_authoring
cpp_data_structures

Big Data Storage and Processing
6 ECTS
big_data_processing_modes
hadoop_big_data
parquet_columnar_storage
cloud_big_data_platforms
web_data_scraping
server_side_php

Edge IoT and Web of Things
6 ECTS
esp32_idf_programming
web_of_things
edge_and_fog_computing
iot_streaming_with_kafka
iot_interoperability_and_constraints
smart_agriculture_iot

Machine-Learning Workloads and Model Optimization
4 ECTS
machine_learning_workload_foundations
machine_learning_optimization_and_parallelization
neural_network_architecture_design
efficient_neural_modeling

Parallel and Accelerated Computing for AI
5 ECTS
distributed_and_hybrid_parallel_programming
cuda_architecture_and_programming
hpc_workload_management
reconfigurable_and_fixed_point_computing
machine_learning_processor_acceleration

Computing History and Service Design
6 ECTS
computing_origins
computing_systems_evolution
operating_system_history
service_structure_mapping
social_implications_of_it
generative_ai_basics

Machine-Learning Platforms and Models
6 ECTS
neural_network_architectures
python_machine_learning_frameworks
generative_model_foundations
cloud_machine_learning_platforms
generative_ai_models
algorithm_selection_and_configuration

Planning and Agent Programming
6 ECTS
multi_agent_artifacts_and_coordination
intentional_agent_programming
strips_planning
advanced_planning_methods
maintenance_analytics
reinforcement_learning_applications

Digital Systems and Cryptographic Arithmetic
6 ECTS
language_syntax_grammars
binary_number_operations
programmable_logic_families
chip_hardware_context
medical_artificial_intelligence
medical_3d_modeling_pipeline

Medical Technology and 3D Printing
6 ECTS
medical_3d_printing
medical_artificial_intelligence
medical_3d_modeling_pipeline
digital_fabrication_and_3d_printing
computer_aided_design
digital_graphics_foundations

Network Analysis and Testing
6 ECTS
large_network_analysis
network_hypothesis_testing
network_analysis_foundations
network_level_metrics
network_node_metrics
network_visualization

Web Standards and Cryptographic Arithmetic
6 ECTS
web_standards
cryptographic_modular_arithmetic
web_data_scraping
internet_applications
server_side_php
computer_instruction_execution

Project Risk Management
1 ECTS
project_risk_management

Topic colour: first scope, in this order, among the
current courses teaching it (by teachers' and programmes' department)
service: some DISI teacher, no DISI degree programme
external: no DISI teacher, no DISI degree programme
borrow: no DISI teacher, some DISI degree programme
weak_internal: some (not all) DISI teachers and DISI degree programmes
internal: only DISI teachers and DISI degree programmes